

Session-10: Stream API & Collection Processing –

10a: Java Deque –

Program:

```
import java.util.*;

public class test {

    public static void main(String[] args) {

        Scanner in = new Scanner(System.in);

        Deque<Integer> deque = new ArrayDeque<>();

        Map<Integer, Integer> freq = new HashMap<>();

        int n = in.nextInt();

        int m = in.nextInt();

        int maxUnique = 0;

        for (int i = 0; i < n; i++) {

            int num = in.nextInt();

            deque.add(num);

            freq.put(num, freq.getOrDefault(num, 0) + 1);

            if (deque.size() > m) {

                int removed = deque.poll();

                freq.put(removed, freq.get(removed) - 1);

                if (freq.get(removed) == 0) {

                    freq.remove(removed);

                }

            }

            if (deque.size() == m) {

                maxUnique = Math.max(maxUnique, freq.size());

            }

        }

        System.out.println(maxUnique);

    }

}
```

Output:

Congratulations

You solved this challenge. Would you like to challenge your friends?  

[Next Challenge](#)

✓ Test case 0

✓ Test case 1 

✓ Test case 2 

✓ Test case 3 

✓ Test case 4 

✓ Test case 5 

✓ Test case 6 

Compiler Message

Success

Input (stdin)

1	6 3
2	5 3 5 2 3 2

Expected Output

1	3
---	---

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10b: Java Hashset –

Program:

```
import java.io.*;
import java.util.*;
import java.text.*;
import java.math.*;
import java.util.regex.*;

public class Solution {

    public static void main(String[] args) {

        Scanner s = new Scanner(System.in);

        int t = s.nextInt();

        String [] pair_left = new String[t];

        String [] pair_right = new String[t];
```

```

for (int i = 0; i < t; i++) {
    pair_left[i] = s.next();
    pair_right[i] = s.next();
}

HashSet<String> set = new HashSet<>();
for (int i = 0; i < t; i++) {
    String pair = pair_left[i] + " " + pair_right[i];
    set.add(pair);
    System.out.println(set.size());
}
s.close();
}
}

```

Output :

Test Case	Status	Input (stdin)	Download
Test case 1	✓	1 1000	Download
Test case 2	✓	2 ae ec	
Test case 3	✓	3 jl jj	
Test case 4	✓	4 gj ja	
		5 mc de	
		6 jc lj	
		7 ca mg	
		8 ni kg	